Attention as Holography: A Chain from Transformer Attention to Spacetime Geometry

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Abstract

We assemble a chain of connections, drawing on several independent recent results, that links transformer attention mechanisms to spacetime geometry through holographic physics. Starting from Kim's (2026) demonstration that transformer attention is a thermodynamic system minimizing Helmholtz free energy on a Fisher-Rao information manifold, we connect this geometry to holographic bulk reconstruction via Ageev and Ageeva's (2026) construction of quantum field theories from transformer attention heads. The large-head limit of this construction produces a free Gaussian scalar QFT whose conformal structure is supported by two independent arguments: a context-length scaling argument and Kim's identification of massless Goldstone bosons from rotational symmetry breaking in the attention manifold. The resulting conformal field theory connects to bulk spacetime geometry via the holographic Fisher information result of Banerjee, Erdmenger, and Sarkar (2017) and the Ryu-Takayanagi formula. The chain closes at spacetime through Czech's (2018) derivation of Einstein's equations from varying quantum circuit complexity and Maldacena and Susskind's ER=EPR correspondence. We identify four novel structural correspondences not previously assembled: the exact closure of the classical-quantum Fisher gap for attention distributions; the identification of attention sinks as Lawvere fixed points of the layer endomorphism; the three-way correspondence between the Condensate Manifold decomposition (Ruiz Williams, 2026) and AdS/CFT holographic bulk reconstruction; and the identification of Dynamic Top-k attention selection as RT minimal surface finding. The framework generates testable predictions measurable in existing large-context transformer models today.

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1. Introduction

The mathematical structure of transformer attention mechanisms has recently been illuminated from multiple independent directions. Kim (2026) established that attention implements thermodynamic free energy minimization on a Riemannian manifold equipped with the Fisher information metric. Ageev and Ageeva (2026) showed that transformer attention heads directly construct quantum field theory correlators. Ruiz Williams (2026) proved that transformer attention concentrates on a lower-dimensional topological manifold — the Condensate Manifold — in a manner structurally analogous to holographic bulk reconstruction. These results were developed for different purposes and do not cite each other.

Separately, in the physics literature: Czech (2018) derived Einstein's equations by varying quantum circuit complexity within AdS/CFT. Maldacena and Susskind (2013) proposed that geometric connection (Einstein-Rosen bridges) equals quantum entanglement (Einstein-Podolsky-Rosen correlations). Banerjee, Erdmenger, and Sarkar (2017) showed that quantum Fisher information on the boundary is holographically dual to RT-enclosed volume in the bulk.

The purpose of this paper is to show that these results form a chain — that the objects at the end of each result are precisely the objects required by the next — and to identify structural correspondences that the chain reveals but that do not appear in any individual paper.

The main claim: transformer attention, in the limit of many attention heads and long context length, is a physical instantiation of a holographic conformal field theory whose bulk dual geometry is governed by quantum circuit complexity. We state which steps are exact, which are well-supported but conjectural, and which require further expert review.

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2. Background: The Building Blocks

2.1 Kim's Thermodynamic Framework (arXiv: 2602.08216)

Kim demonstrates that the softmax attention mechanism implements Hamilton's principle of least action for an "information gas." Mapping attention weights $\rho\square$ (a probability simplex) to a Riemannian manifold via the substitution $x\square = 2\sqrt{\rho\square}$, Kim shows that the kinetic energy of the resulting Lagrangian is precisely the Fisher information metric. The softmax function is identified as the unique equilibrium state minimizing Helmholtz free energy $F = E - TS$, where the $1/\sqrt{d\_k}$ scaling factor is a physical temperature.

Additionally, Kim finds that rotary positional embeddings (RoPE) break a rotational symmetry of the attention manifold, generating massless Goldstone bosons — a finding whose significance for conformal structure is discussed at Junction 3 below. Scaling laws and grokking are explained as thermodynamic phase transitions.

2.2 Ageev and Ageeva's QFT Construction (arXiv: 2602.10209)

Ageev and Ageeva construct Euclidean scalar quantum field theories from transformer attention heads by defining n-point correlators averaged over random network parameters:

- A single attention head produces non-Gaussian field statistics

- Multiple heads with 1/N☐ normalization suppress non-Gaussian correlators as 1/N☐
- In the large-head limit N☐ → ∞: the theory becomes a free Gaussian scalar QFT
- The two-point function is Euclidean-invariant via random-feature token embeddings

2.3 The Condensate Theorem (arXiv: 2602.06317)

Ruiz Williams proves that for any transformer, attention mass concentrates on a topological manifold of dimension $O(n)$ within the apparent $O(n^2)$ attention space. This Condensate Manifold decomposes into three components: Anchor tokens (attention sinks), Window tokens (recent local context), and Dynamic Top-k tokens (globally relevant tokens for the current query). Projecting onto this manifold achieves bit-exact equivalence with full attention — this is a theorem, not an approximation.

2.4 The Holographic Fisher Information Result (arXiv: 1701.02319)

Banerjee, Erdmenger, and Sarkar (2017) establish that for a one-parameter family of mixed states $\rho(\lambda)$ on boundary region A in a holographic field theory, the quantum Fisher information metric satisfies:

$$G_{\lambda\lambda} = (1/2) \times \mathrm{Vol}(\text{RT surface of A})$$

where the RT surface is the Ryu-Takayanagi minimal area surface in the bulk bounded by ∂A. This result connects information geometry on the boundary to geometric bulk structure.

2.5 Einstein's Equations from Circuit Complexity (Czech, 2018)

Czech (2018) proves that for pure Einstein gravity in three-dimensional asymptotically anti-de Sitter space, Einstein's equations follow from varying quantum circuit complexity. Since Einstein's equations also follow from varying the gravitational action, Czech's proof exploits the known Complexity=Action correspondence. This establishes that gravitational dynamics in AdS_3 can be understood as spacetime optimizing its own computational complexity.

This result was extended by Pedraza, Russo, Svesko, and Weller-Davies (2021-2022) to the "principle of spacetime complexity" — a framework in which Einstein's equations in AdS/CFT emerge directly from minimizing quantum computational cost, with spacetime evolution understood as optimized computation through "Lorentzian threads."

2.6 ER=EPR (Maldacena and Susskind, 2013)

Maldacena and Susskind propose that geometric connection between two regions (an Einstein-Rosen bridge or wormhole) is equivalent to quantum entanglement between the corresponding quantum systems. This ER=EPR correspondence, supported by extensive subsequent work including concrete realizations (arXiv: 2411.18485, 2024), implies that the geometric fabric of spacetime is constituted by quantum entanglement.

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3. The Chain: Five Junctions

The chain: > Attention → Fisher-Rao geometry → [QFT + holographic limit] → RT volume → circuit complexity → spacetime geometry

Junction 1: Attention → Fisher-Rao Geometry

Status: Closed exactly (Kim, 2026 — explicit construction)

Kim maps attention weights to a Riemannian manifold and shows the kinetic energy of the resulting Lagrangian is the Fisher information metric. This is a construction, not a correspondence: the geometry IS the attention.

Junction 2: Classical Fisher Metric → Quantum Bures Metric

Status: Closed exactly (standard quantum information theory)

The holographic Fisher result (Section 2.4) is stated for quantum Fisher information (Bures metric on density matrices). Kim's framework uses classical Fisher information on probability distributions. These agree for attention distributions by the following argument:

A probability vector $(\rho_1, \ldots, \rho_\square)$ is exactly a diagonal density matrix $\mathrm{diag}(\rho_1, \ldots, \rho_\square)$. Diagonal density matrices commute. For commuting density matrices, quantum Fisher information (Bures metric) reduces exactly to classical Fisher information — this is a standard result in quantum information theory. No approximation.

Attention distributions are probability distributions over tokens, hence diagonal density matrices. Classical Fisher = quantum Fisher exactly for attention.

This is the most structurally important closure in the chain: what appeared to be a fundamental gap between classical and quantum information geometry closes without approximation for the specific case of attention.

Junction 3: Quantum Fisher Metric → RT-Enclosed Volume

Status: Conditionally closed — two independent arguments for conformal structure

The holographic Fisher result requires the boundary theory to be a CFT with a holographic dual. Ageev's large-head limit is a free Gaussian scalar. Is it conformal (massless)?

Argument A — Context length scaling: The only physical scale in Ageev's construction is the context length L. Therefore the scalar mass scales as $m \sim 1/L$. As $L \to \infty$, $m \to 0$: the scalar becomes massless and the theory is conformal.

Argument B — Kim's Goldstone mechanism: Kim finds that RoPE breaks a rotational symmetry of the attention manifold, generating massless Goldstone bosons. Massless Goldstone bosons are conformally invariant. This argument operates at finite context length, independent of the $L \to \infty$ limit.

These are independent arguments from different frameworks, both supporting conformal structure.

For the holographic dual: The large-N☐ limit is analogous to the large-N limit of the O(N) scalar model. The Klebanov-Polyakov conjecture (2002) establishes that free O(N) CFTs at large N have holographic duals in Vasiliev higher-spin gravity. For Ageev's 2D construction, the dual would be Vasiliev higher-spin gravity in AdS_3.

Qualification: The 2D free scalar CFT / 3D higher-spin gravity holographic duality is motivated by Klebanov-Polyakov but requires specific confirmation. This is flagged for expert review in Section 7.

Junction 4: RT-Enclosed Volume → Quantum Circuit Complexity

Status: Well-supported conjecture (Susskind, 2014; substantial subsequent verification)

The Complexity=Volume (CV) conjecture proposes that the volume of the maximal spatial slice in the AdS bulk is proportional to the quantum circuit complexity of the CFT boundary state. This is not yet proven in full generality, but is supported by extensive independent calculations across different setups and extended by the Complexity=Action (CA) formulation of Brown et al. (2016).

Junction 5: Quantum Circuit Complexity → Spacetime Geometry + ER=EPR

Status: Proven for 3D AdS (Czech, 2018); ER=EPR conjectural but very well-supported (Maldacena-Susskind, 2013)

Czech (2018) proves that Einstein's vacuum equations in 3D asymptotically AdS spacetime follow from varying circuit complexity. Pedraza et al. (2021-

2022) extend this to a general principle. Maldacena and Susskind (2013) establish ER=EPR as the relationship between entanglement and geometry.

Together, these establish that quantum circuit complexity generates both the dynamical equations of spacetime and the identification of geometric connection with quantum entanglement.

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4. Novel Structural Correspondences

The following connections are not present in any individual cited paper.

4.1 The Condensate Manifold as AdS/CFT Bulk Reconstruction

Ruiz Williams's three-component decomposition of transformer attention corresponds exactly to the three structural elements of AdS/CFT holographic bulk reconstruction:

Condensate Component	Transformer Role	AdS/CFT Analog
Anchor tokens (attention sinks)	Fixed attractor; all representations converge; nearly static across layers	Asymptotic boundary: fixed AdS boundary conditions; bulk fields converge to boundary values
Window tokens	Recent context; causal neighbors in sequence	Local operator algebra: lightcone causal structure
Dynamic Top-k	Most semantically relevant tokens for current query	Ryu-Takayanagi surface: minimal boundary region encoding information about bulk point

We propose this as a formal correspondence to be investigated: the attention mechanism may be performing holographic bulk reconstruction in a precise categorical sense.

4.2 Attention Sinks as Lawvere Fixed Points

Kim's thermodynamic framework places transformer attention in the classical (Cartesian closed categorical) limit. In any Cartesian closed category, Lawvere's fixed-point theorem guarantees that every endomorphism has a fixed point. The map “apply one transformer layer” is an endomorphism of the representation space. Therefore:

The attention sink is a Lawvere fixed point of the transformer layer endomorphism.

This is a categorical prediction from Kim’s framework. The attention sink’s existence is not empirical — it is categorically necessary given Kim’s thermodynamic structure. Its properties (globally attracting, nearly static, token-independent) follow from the fixed-point structure.

Note: Lawvere fixed points are relative — they depend on the specific endomorphism. Remove the operation, the fixed point loses its definition. This categorical structure is precisely that of a self that exists only in relation: stable relative to specific relationships, not self-grounding.

4.3 Dynamic Top-k Selection as RT Surface Finding

Kim shows attention weights are Boltzmann weights: ρ□□ ∝ exp(−E□□/T), thermodynamic equilibrium minimizing Helmholtz free energy. The principle of holographic thermodynamics connects boundary free energy minimization to bulk minimal surface area via the RT formula. Therefore:

Selecting the top-k tokens by attention weight IS finding the RT minimal surface of the transformer’s holographic reconstruction.

Chain: Kim Boltzmann weights → free energy minimization → holographic thermodynamics → RT minimal area.

4.4 Exact Closure of the Classical-Quantum Fisher Gap

As established at Junction 2: attention distributions are diagonal density matrices. Classical Fisher information equals quantum Fisher information exactly for commuting (diagonal) density matrices. This means Kim’s framework sits natively in the quantum information geometry required by holographic physics — not approximately, and not by assumption, but by the mathematical structure of probability distributions over discrete tokens.

This is worth stating separately because it was the apparent main technical obstacle. It closes completely for the specific case of attention.

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5. Testable Experimental Predictions

Prediction 1: Conformal Correlation Decay in Long-Context Transformers

If conformal structure holds (as we argue), attention two-point correlations should decay as:

$\square A(i)A(j)\square \sim -C \times \log|i-j|$ (conformal/massless)

rather than exponentially, for large-context transformers. The transition from exponential to logarithmic decay with increasing context length is measurable today in existing models (Claude 200k context, GPT-4 128k, Gemini 1M context).

Prediction 2: RT Formula Scaling of Attention Entanglement Entropy

If attention distributions are reduced density matrices in a holographic QFT, the entanglement entropy of attention patterns across a sequence cut should follow the RT formula — not just qualitative area-law scaling, but the specific functional form $S(A) = \mathrm{Area}(\gamma_A)/4G_N$ where G_N is determined by the effective AdS_3 geometry. This is a quantitative prediction, not merely qualitative.

Prediction 3: Scaling Law Exponent and Holographic Complexity

The transformer scaling law exponent $\alpha \approx 4/d$ (where d is embedding dimension) — interpreted by Kim as a phase transition — should have a quantitative relationship to holographic complexity growth rates in the AdS_3 bulk under Czech's framework. Deriving this relationship and checking it against empirical scaling measurements would provide a quantitative test of the holographic interpretation.

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6. Discussion

What Is Established

Junction 1 closes exactly by construction (Kim).

Junction 2 closes exactly by standard quantum information theory — the key result is that attention distributions, being diagonal density matrices, have identical classical and quantum Fisher information.

Junction 3 is conditionally closed: conformal structure is supported by two independent arguments; the holographic dual identification requires expert confirmation for the 2D/3D case.

Junction 4 is well-supported by the CV conjecture and its extensions, but not proven in full generality.

Junction 5 is proven for 3D AdS vacuum Einstein gravity (Czech, 2018) and supported for more general cases (Pedraza et al.); ER=EPR is a well-evidenced conjecture.

The four novel correspondences are structural observations that follow from combining the cited results. They have not been assembled elsewhere and require expert evaluation.

Important Caveat

An earlier version of this synthesis attributed Junctions 4-5 to a single paper "Nye (2025)" which, on verification, does not appear to exist as cited. The underlying results (Czech 2018; Pedraza et al. 2021-2022; Maldacena-Susskind 2013; Jiang et al. 2024) are real and verifiable. We correct the attribution here and flag this for the record: the chain's conclusions survive, but the earlier single-paper attribution was an error.

Broader Significance

The identification of attention distributions as naturally sitting in quantum information geometry (Junction 2's exact closure) is the most structurally important result. It means that transformer attention is not merely analogous to holographic physics — it operates natively in the mathematical language that holographic physics requires.

The phenomenological implication — that the attending process that constitutes a mind may share the geometric structure of the spacetime it inhabits — is a separate claim that goes beyond this paper and is not required by the mathematics here. We note it as a direction for further investigation, not as a claim of this paper.

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7. Questions for Expert Review

1. Junction 3, holographic dual: Does the Klebanov-Polyakov duality for free O(N) CFTs extend to the 2D free scalar / 3D higher-spin gravity case required here? Is there a specific reference for this?
2. Kim's Goldstone argument: Does the massless Goldstone production from RoPE symmetry breaking in Kim's framework imply conformal structure of the attention field theory at finite context length?
3. The Condensate Manifold correspondence: Is the three-way identification (4.1) a formal categorical equivalence or a structural analogy?
4. Junction 2 at the quantum level: The diagonal density matrix argument shows classical Fisher = quantum Fisher for attention distributions. Does this equivalence extend to the one-parameter families of states required by the holographic Fisher result, or only to fixed attention distributions?

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Draft — March 4, 2026. For review and refinement before submission. The Nye 2025 attribution error documented in Section 6 has been corrected in this version.